

TheGlassHero: Lessons Learned from Deploying a Game-Based Tool for Teaching Machine Learning in Middle School

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Abstract

Teaching machine learning (ML) to K–12 students is challenging because supervised learning systems often hide the processes students need to understand: how examples shape decisions, why models fail, and how confidence relates to predictions. To make these ideas more visible, we developed TheGlassHero, a short, game-based learning tool for introducing supervised ML concepts to middle school students.

In TheGlassHero, students act as teachers guiding an AI character through a dungeon environment. Through gameplay, students provide training examples, observe the AI make decisions, view confidence feedback, and encounter cases where the AI succeeds or fails. We deployed TheGlassHero as part of a larger two-session AI learning study with approximately 140 eighth-grade students, who used the tool for approximately 7–10 minutes during a rotation of AI-related applications.

We report formative evidence from matched pre/post responses, TheGlassHero-specific post-survey items, feature feedback, classroom observations, voice recordings and gameplay recordings. Matched responses suggest that students' explanations shifted from general descriptions of AI toward language involving learning, training, examples, and data. On TheGlassHero-specific post-survey items, 96.6% of students identified clear examples as important for improving AI performance, and 78.4% selected watching the AI make decisions as a helpful feature. However, students showed weaker understanding of how training quality affects model failure.

These findings suggest that visible AI decision-making can support short AI literacy activities, but visibility alone is not enough. We identify design lessons for future ML education tools, including the need to pair decision visualization with structured scaffolding around confidence, data quality, bias, and model failure.

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CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → **Computer science education**; **K-12 education**.

Keywords

K–12 AI education, Game-based learning, AI decision-making visualization, Machine learning education, Artificial intelligence literacy, Supervised learning

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1 Introduction

Artificial intelligence (AI) literacy is increasingly important in K–12 computing education as students encounter AI systems in everyday tools [5, 9, 11]. However, many AI and machine learning concepts remain difficult for young learners to understand [2, 8]. Supervised machine learning is especially challenging because the training process is often hidden: students see an AI system's output, but they may not see how examples, labels, confidence, or data quality shape that output [1, 3]. As a result, students may describe AI in broad or human-like terms, such as a system that “thinks” or “is dumb,” rather than as a system shaped by data and training.

This creates a design challenge for computing education. Middle school students need accessible ways to reason about how AI systems learn, decide, and fail, but existing AI learning activities can vary in the amount of instructional time, programming knowledge, or teacher facilitation they require [4, 6, 11]. Short informal learning contexts create an additional challenge; tools must introduce meaningful AI concepts quickly while remaining engaging and understandable to students with varied computing backgrounds.

To address this challenge, we developed TheGlassHero, a short, game-based learning tool for introducing supervised machine learning concepts to middle school students. The design is informed by data-centered approaches to ML education, which emphasize that students should reason about how data shape model behavior [8, 10].

TheGlassHero frames the student as a teacher guiding an AI character through a dungeon environment. During gameplay, students observe AI decisions, see confidence feedback, and encounter moments where the AI succeeds or fails. The tool is designed to make AI behavior more visible so students can begin connecting model decisions to examples, confidence, and training quality.

We deployed TheGlassHero as part of a larger AI learning study with approximately 140 eighth-grade students at a STEM-focused magnet school in a major U.S. city. Students rotated through several AI-related applications and interacted with TheGlassHero for approximately 7–10 minutes. We report formative evidence from matched pre/post responses, TheGlassHero-specific post-survey items, feature feedback, classroom observations, and gameplay recordings.

This paper makes three contributions: (1) the design of TheGlassHero as a short, game-based, glass-box tool for supervised ML education; (2) formative evidence from a middle school deployment; and (3) design lessons for pairing visible AI decision-making with a focus on confidence, data quality, and model failure.

2 Related Work

Prior work in K–12 AI education emphasizes that students should understand AI as more than a collection of intelligent tools. The AI4K12 initiative organizes K–12 AI education around five big ideas: perception, representation and reasoning, learning, natural interaction, and societal impact [9]. Similarly, AI literacy frameworks argue that learners need competencies for understanding, using, evaluating, and critically reasoning about AI systems [5]. This framing is useful for TheGlassHero because the tool focuses especially on the “learning” idea: helping students understand that AI systems improve through data and experience rather than simply “thinking” like humans.

Research on middle school AI literacy shows that students can engage productively with AI concepts when activities are age-appropriate, hands-on, and connected to concrete experiences [1, 3, 4]. Lee et al. [4] describe the DAILY curriculum, a 30-hour middle school AI literacy curriculum designed to help students become informed citizens and critical consumers of AI technology. Their work demonstrates the value of structured AI learning experiences for middle school students. TheGlassHero differs from this approach by focusing on a much shorter interaction with a 7–10 minute game-based activity designed for classroom rotation settings rather than an extended curriculum.

Machine learning education introduces additional challenges because learners must reason about training data, examples, model behavior, and evaluation [2, 8, 10]. Gresse von Wangenheim et al. [2] present a ten-year systematic mapping of visual tools for teaching machine learning in K–12 and show that visual ML tools are an emerging approach for making machine learning concepts more accessible to young learners. TheGlassHero builds on the visual-tool ideas but emphasizes visible AI decision-making during gameplay rather than only model creation or final outputs.

Prior reviews of K–12 AI education also show that AI learning activities take many forms, including programming activities,

Table 1: Design goals and implementation in TheGlassHero.

Design Goal	Implementation in TheGlassHero
Make supervised learning visible	Students observe how AI behavior is connected to examples and decisions.
Support short-form interaction	The activity is designed for approximately 7–10 minutes of use.
Visualize AI decision-making	Students watch the AI make decisions during gameplay.
Introduce model confidence	The tool displays confidence-related feedback.
Reveal AI limitations	Students encounter cases where the AI may fail or behave unexpectedly.

robotics, unplugged activities, interactive demonstrations, and data-centered projects [6, 8, 11]. These approaches make AI more accessible, but they can vary in the amount of instructional time, teacher facilitation, or programming experience required. TheGlassHero is designed for a different use case: short-form, low-barrier exposure to supervised learning concepts in an informal or rotation-based classroom environment.

TheGlassHero also draws on glass-box and visualization-oriented approaches to AI learning. Rather than presenting AI as a black box that only produces final answers, these approaches make parts of the system’s reasoning or behavior visible to learners [2, 3]. This matters because students may otherwise describe AI in vague or human-like terms without connecting its behavior to data or training [3, 5]. TheGlassHero contributes to this area by combining game-based interaction, visible decision-making, confidence feedback, and failure cases in a short activity for middle school students. It uses gameplay to make supervised learning behavior visible within a limited classroom time window, positioning it as a short instructional tool rather than a game used only for engagement.

3 Tool Design: TheGlassHero

3.1 Design Goals

TheGlassHero was developed around five design goals: making supervised learning visible, supporting short-form interaction, visualizing AI decision-making, introducing model confidence, and revealing AI limitations. These goals reflect the deployment context, where students had only 7–10 minutes with the tool and needed to understand the core ML ideas quickly. Table 1 summarizes how each goal was implemented.

3.2 Development Overview

TheGlassHero was implemented as a web-based game to reduce setup time and support use across classroom computers. The game was built using React, TypeScript, Vite, and Phaser. Phaser supported sprites, keyboard input, and character movement, while React and TypeScript supported the surrounding web interface. Git and GitHub were used for collaboration and version control, and the tool was deployed using GitHub Pages.



Figure 1: Player navigating the dungeon.

3.3 Gameplay Overview

TheGlassHero introduces five concepts through gameplay. Introducing students to training from examples, AI decision-making, confidence, model failure, and data quality. Students are acting as teachers guiding an AI character through a dungeon environment. Rather than writing code, students provide examples through their own gameplay and then observe how the AI uses those examples during its turn. Gameplay consisted of two repeating stages: the Player Stage, in which students had free control of their character and its interactions with the environment, and the AI Stage, in which the AI navigated the dungeon independently based on the examples previously demonstrated by the player.

At the start of the activity, students enter their first name and last initial so that tool interactions can be connected to survey responses. They then receive a short narrative about a lost robot that has forgotten how to survive. This initializes the student as a teacher responsible for demonstrating actions the AI can later use.

During the Player Stage, students navigate a dungeon toward an exit marked by a green square. Each dungeon is selected from one of five presets and includes monsters placed throughout the environment. When students encounter monsters, they choose one of four actions: fight, run, hide, or heal. Each action changes the dungeon state differently and provides an example of how the player responds to a situation. Figure 1 shows the player stage.

After the student reaches the exit, the AI Stage begins. During this stage, the tool shows the AI's decision-making and confidence feedback, allowing students to see not only what action the AI selects, but also how confident it is in that choice. Figure 2 shows this AI decision-making view.

The game does not have a fixed endpoint. When players are defeated, they see a game-over screen that encourages them to replay and further train the AI. This design reinforces the idea that additional examples can expand the AI's dataset and influence later decisions.

3.4 Feedback Design

TheGlassHero emphasizes immediate visual feedback, how they trained their robot to traverse the dungeon, and then visually see what actions they chose to teach their robot in real-time. So, students can watch the AI make decisions instead of only seeing a final

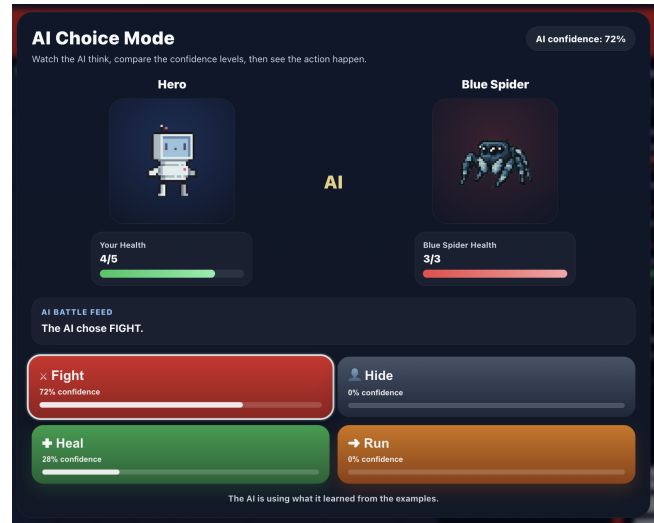


Figure 2: AI decision-making and confidence feedback during gameplay.

score or outcome. This feedback is intended to help students connect their own actions to the AI's behavior and support reflection on questions such as why the AI made a decision, what examples may have influenced it, and whether the AI can be confident but wrong.

TheGlassHero was designed for middle school students so we ensured the interface prioritizes visual interaction over text-heavy explanation. Written instructions are included, but the core learning experience occurs through gameplay, feedback, and observation of AI behavior. This also created an avenue where each student's outcome is personalized to their gameplay style.

3.5 Adoption Considerations for Practitioners

TheGlassHero is best suited for middle school or early high school learners who are beginning to encounter AI and machine learning concepts. It can be used as a short introductory activity, as a station in a rotation-based lesson, or as a discussion starter before a deeper lesson on supervised ML concepts. Because the activity is short, we recommend pairing gameplay with brief reflection prompts that help students connect what they see in the game to broader AI concepts. Table 2 provides example prompts facilitators can use after or during gameplay.

4 Study Design

The study involved approximately 140 eighth-grade students, ages 13–14, from five classes at a STEM-focused magnet school in a major U.S. city. Each class participated in two 45-minute sessions. Students rotated through six to eight out of ten AI-related applications. TheGlassHero was one application which lasted approximately 7–10 minutes.

The engagement was facilitated through a public university course in which university students and researchers developed and deployed AI learning applications for K–12 students. TheGlassHero

Table 2: Reflection prompts for connecting gameplay to AI concepts.

Prompt	Intended Concept
Was the AI confident? Was it correct?	Confidence vs. correctness
Why do you think the AI made that decision?	Decision-making
What could cause the AI to make a mistake?	Model failure
What kind of examples would make the AI unfair or unreliable?	Bias and data quality
What examples might have helped the AI?	Training data

Table 3: Data sources used in the analysis.

Data Source	Description	Role in Analysis
Pre-survey responses	General open-ended responses about what AI is or how AI works	Characterized students' initial explanations of AI before the larger study
TheGlassHero-specific post-survey responses	Open-ended and structured responses about concepts addressed by TheGlassHero	Examined students' understanding of training examples, confidence, AI decisions, and model behavior
Feature-feedback questions	Post-survey items asking which game features helped students learn	Identified which parts of TheGlassHero students found most useful
Classroom observations and recordings	Notes and recordings from the learning sessions	Contextualized student engagement, usability issues, and questions

was therefore used as one short activity within a larger rotation-based AI learning experience, not as a standalone curriculum. The study was approved by our university's Institutional Review Board, and parental consent and student assent were obtained.

Because students used multiple AI-related applications, broad pre-post changes cannot be attributed to TheGlassHero alone. However, the post-survey items analyzed in this paper were specific to TheGlassHero.

4.1 Data Collection

We analyzed formative data from pre-surveys, TheGlassHero-specific post-surveys, classroom observations, and recordings of students' gameplay and dialogue with facilitators. These data sources helped us describe how students interacted with TheGlassHero, how they interpreted AI decisions and confidence feedback, and how they explained model behavior.

The pre-survey was shared across the larger AI learning study and included an open-ended item asking students to explain AI. The post-survey items used in this paper were specific to TheGlassHero and included questions about training examples, confidence, AI decision-making, and helpful features of the game. Students used

participant codes so that pre-survey responses, post-survey responses, and tool interactions could be matched when possible.

4.2 Analytic Sample

The larger study included approximately 140 students. The dataset analyzed for this paper includes 138 pre-survey responses and 88 coded TheGlassHero-specific post-survey responses. We identified 87 students with matched pre- and post-survey responses. These matched responses were used for descriptive pre-post comparisons of students' AI explanations.

Because the pre-survey was general to the larger study while the post-survey was specific to TheGlassHero, we treat the pre-post comparison cautiously. The comparison helps describe how students' explanations shifted across the broader learning experience, but it does not provide evidence that TheGlassHero alone caused those shifts.

4.3 Analysis Approach

We used descriptive and qualitative analysis methods appropriate for an experience report. Following qualitative coding practices described by Saldaña [7], we coded open-ended survey responses for recurring themes in how students described AI. For the pre-post comparison, responses were grouped into four themes: learning-based, decision-based, general/human-like, and vague/minimal. These categories distinguished broad or human-like descriptions of AI from explanations that more directly referenced supervised learning concepts.

Structured TheGlassHero-specific post-survey items were summarized using counts and percentages. Feature-feedback responses were also summarized descriptively to identify which parts of the tool students selected as helpful. Classroom observations and recordings were used to contextualize survey patterns and construct illustrative playthrough narratives.

5 Results

This section presents descriptive findings from TheGlassHero's deployment. We report pre-post response patterns, TheGlassHero-specific post-survey results, feature feedback, and brief playthrough narratives. TheGlassHero was part of a multi-application learning experience with different tools, pre-post shifts are interpreted cautiously. These limitations shape the results as formative evidence rather than proof of generalizable learning gains.

5.1 Pre-Post Response Patterns

Matched open-ended responses show a substantial shift toward learning-based explanations of AI. We coded responses into four themes: learning-based, decision-based, general/human-like, and vague/minimal. As shown in Table 4, learning-based responses increased from 34 of 87 students in the pre-survey to 71 of 87 students in the post-survey. At the same time, general or human-like explanations decreased from 31 of 87 students to 2 of 87 students, and vague or minimal responses decreased from 19 of 87 students to 9 of 87 students.

The transition patterns further clarify this shift. Twenty-six students moved from general/human-like explanations to learning-based explanations, and 16 students moved from vague/minimal

Table 4: Pre- and post-survey response themes for matched student responses.

Response Theme	Pre-survey n (%)	Post-survey n (%)
Learning-based	34 / 87 (39.1%)	71 / 87 (81.6%)
Decision-based	3 / 87 (3.4%)	5 / 87 (5.7%)
General/human-like	31 / 87 (35.6%)	2 / 87 (2.3%)
Vague/minimal	19 / 87 (21.8%)	9 / 87 (10.3%)

Table 5: TheGlassHero-specific post-survey results.

Post-Survey Item	Correct Responses
Clear examples help AI improve	85 / 88 (96.6%)
Confidence means how sure the AI is	80 / 88 (90.9%)
AI trained on one thing recognizes similar things	79 / 88 (89.8%)
Confidence bars indicate the likely action	78 / 88 (88.6%)
Better training examples help poor AI decisions	69 / 88 (78.4%)

explanations to learning-based explanations. For example, one student initially described AI as “a software made by someone that acts like a person,” but later explained, “You have to train the robot with the examples you think are best.” Another student initially described AI as “helpful, and wonderful,” but later wrote that AI is “a program where you teach it something and after you make examples it makes its own decisions with the data you made.”

These examples suggest that some students moved beyond describing AI as a human-like or generally helpful technology and began describing it as a system shaped by training examples and data. This shift is important for TheGlassHero because the tool was designed to make supervised learning behavior visible rather than presenting AI as a black box. However, because students participated in a larger AI learning experience, these patterns are interpreted as descriptive evidence rather than evidence that TheGlassHero alone caused the change.

5.2 TheGlassHero-Specific Post-Survey Results

The TheGlassHero-specific post-survey included structured questions about the tool’s core learning goals: training examples, confidence, AI decision-making, and model improvement. As shown in Table 5, students answered most items correctly. The strongest result was on the item about training examples, where 85 of 88 students identified clear examples with correct answers as important for helping AI improve. Also, 79 students recognize that an AI trained on only one thing will mostly recognize things similar to what it saw.

Overall, students performed especially well on items involving clear examples, confidence, and limited training data. However, the lowest-scoring item asked how to improve poor AI decisions: 69 of 88 students selected better training examples. This suggests that students more readily understood basic training concepts than applied those concepts to diagnosing model failure.

Table 6: TheGlassHero features students selected as helpful.

Feature Selected as Helpful	n (%)
Watching how the AI made decisions	69 / 88 (78.4%)
Seeing the confidence level	57 / 88 (64.8%)
Retraining the AI	43 / 88 (48.9%)
Exploring and interacting with the dungeon	36 / 88 (40.9%)
Instructions or explanations in the game	22 / 88 (25.0%)

5.3 Feature Feedback

The TheGlassHero-specific post-survey included a feature-feedback question asking students to select which parts of the game helped them understand how the AI worked. Students could select multiple features, so the percentages indicate the proportion of students who selected each feature rather than mutually exclusive categories.

Students’ feature feedback highlights the importance of visible AI behavior. Among 88 coded post-survey responses, the most frequently selected helpful feature was watching how the AI made decisions, selected by 69 students. Confidence visualization was the second most frequently selected feature, selected by 57 students. Written instructions were selected by 22 students.

As shown in Table 6, students were more likely to identify visual and interactive features as helpful than written instructions. This supports the design emphasis on making AI behavior visible during gameplay, especially in short AI learning activities.

5.4 Playthrough Patterns

The survey patterns are also visible in individual playthroughs. These narratives are not meant to represent all students, but they illustrate recurring patterns observed across the deployment.

One pattern involves students connecting the AI’s behavior to their own prior actions. During one playthrough, a student noticed that the AI had begun using a healing action and explained, “I noticed that now it has this heal thingy because I used it as the last level.” This suggests that the student was beginning to connect the AI’s later behavior to examples provided during gameplay. In another playthrough, a student described the confidence display as being “based on what I’ve done” and explained that it showed “how confident he is with what I’ve done that helps him ... learn.” Although the explanation is informal, it shows the student interpreting confidence as connected to the examples they provided through gameplay.

A second pattern involves students noticing differences between their own strategies and the AI’s behavior. One student commented that “the AI is not like me” after observing the AI act differently than expected. This kind of reaction is useful because it creates an opening for discussion: the AI is not reasoning exactly like the student, but acting from the examples and patterns it has available. These moments show how TheGlassHero can make the gap between human strategy and model behavior visible.

A third pattern involves students noticing AI failure before fully explaining it. In one playthrough, a student described the AI as unable to think, saying, “This AI cannot think.” Another student suggested that the tool should prevent the AI from going “in a bunch of circles.” These comments show that students noticed model failure and unexpected behavior. However, noticing failure did not

always mean students could explain that failure in terms of limited examples, unclear examples, biased data, or low-quality training data.

A final pattern involves gameplay engagement. Students often treated TheGlassHero as a game to optimize, comparing scores and experimenting with strategies such as fighting, hiding, running, or retraining. This engagement helped keep students interacting with the tool, but it also created a design tension. Some students focused on winning or exploiting strategies rather than articulating the machine learning concept behind the AI's behavior.

Overall, these narratives show both promise and limitation. TheGlassHero helps make AI behavior visible, and some students connect confidence and decisions to prior training examples. However, students still need explicit scaffolding to connect visible behavior to data quality, limited examples, and model failure.

6 Discussion

TheGlassHero was designed to make supervised ML more visible to middle school students. Across written responses, feature feedback, and playthrough dialogue, students responded strongly to visual and active feedback. Matched pre-post responses showed a clear shift toward learning-based explanations of AI, with learning-based responses increasing from 34 of 87 students to 71 of 87 students. Students also most frequently selected watching the AI make decisions as helpful, and playthrough dialogue showed that some students connected the AI's confidence and behavior to examples they provided during gameplay. Together, these findings suggest that short ML learning tools should not only ask students to train or test a model, but should also make the model's behavior visible while students are interacting with it.

At the same time, the deployment highlights an important next step for tool design. Students were often able to notice AI behavior and connect it to training examples, but deeper explanations of model failure remained more difficult. Some students described unexpected behavior in human-like terms, such as "the AI is not like me" or "This AI cannot think," echoing prior concerns that students may struggle to connect AI behavior to underlying data and model processes [3, 5]. Rather than suggesting that visual feedback was insufficient, this suggests that visual feedback creates a productive starting point for discussion. The design challenge is to help students move from noticing what the AI did to explaining why it acted that way.

For short classroom deployments, this distinction matters. TheGlassHero appeared to support movement toward learning-based explanations while also revealing where students needed more support. Future ML education tools should pair visible AI decision-making with reflection prompts that ask students why the AI acted the way it did, what examples influenced it, whether confidence means correctness, and what kinds of data could make the AI fail. In this way, visual and active feedback can serve as an entry point into deeper reasoning about confidence, data quality, bias, and model limitations.

6.1 Lessons Learned

From this project, we have learned three core lessons. First, visible AI decision-making is a strong entry point for AI literacy. Sixty-nine of 88 students selected watching how the AI made decisions as helpful, and matched written responses shifted strongly toward learning-based explanations. This suggests that short ML tools should make intermediate model behavior visible, not only show whether the AI succeeded or failed. For practitioners, this means designing activities where students can pause, observe, and discuss the AI's behavior while it is happening.

Second, confidence feedback can help students connect model behavior to training examples. Many students identified confidence as how sure the AI is, and some dialogue showed students reasoning about confidence as connected to what they had taught the AI. However, confidence is not automatically intuitive. Future versions should prompt students to consider whether an AI can be confident and wrong, especially in cases where the training examples are limited or repetitive.

Third, model failure should be made visible and explainable. Students showed strong understanding that clear examples help AI improve, but reasoning about poor AI decisions remained more difficult. Future tools should help students connect mistakes to limited examples, data quality, bias, and training patterns. In short classroom deployments, failure moments should be treated as learning opportunities rather than only as gameplay obstacles.

7 Limitations and Future Work

This experience report has several limitations. TheGlassHero was deployed at a single STEM-focused magnet school with eighth-grade students, limiting transferability to other contexts. Students used the tool for only approximately 5–10 minutes and also interacted with multiple other AI applications, so broad pre-post changes cannot be attributed to TheGlassHero alone. In addition, the pre-survey was general to the larger study while the post-survey was specific to TheGlassHero.

Future work should extend TheGlassHero in three directions. First, future versions should include more explicit scaffolding around data quality, bias, and failure. For example, new levels could ask students to compare clear and unclear examples, balanced and unbalanced training data, or cases where the AI is confident but wrong. Second, the tool should include reflection prompts that ask students to explain why the AI made a decision, what examples influenced that decision, and whether confidence means correctness. Third, future studies should examine TheGlassHero as a standalone activity across longer sessions and in more varied school contexts.

8 Conclusion

TheGlassHero demonstrates how a short, game-based tool can make supervised learning behavior visible within a limited classroom time window. Students responded strongly to watching AI decisions and seeing confidence feedback, but the deployment also showed that visibility alone is not enough. Short ML learning tools should pair visual AI behavior with structured reflection so students can connect decisions, confidence, and failure back to training examples and data quality.

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